

Designing a Study: Sleep, Emotion, Memory, and Language

Throughout this course you've read about experiments that reveal how the brain generates behaviors, emotions, memories, etc. This assignment is an opportunity to design a study testing these experiments yourself. Turning a curiosity into a testable hypothesis, and a hypothesis into a controlled comparison, is a different skill than understanding a study someone else already ran, and it's one of the core habits of scientific thinking. For this final assignment, you will design an original (but realistic) study addressing a question at the intersection of two topics from this unit — sleep, emotion, memory, and language — or connecting one of these topics to material from earlier in the course. You do not need to run the study. The goal is to think like a researcher: to specify what you would manipulate, what you would measure, what results would support or undercut your idea, and why.

Instructions

1. Choose a question: Pick a specific, testable question that sits at the intersection of at least two topics — either two from this unit (sleep, emotion, memory, language) or one from this unit paired with an earlier topic in the course (e.g., neurotransmitters, any of the sensory systems, neurophysiology). The question should be narrow enough that you can imagine an actual experiment testing it, not just a broad topic to describe.

Example Questions:

- Does sleep deprivation impair memory for emotional content?
- Does REM sleep selectively consolidate emotionally arousing memories over neutral ones?
- Does induced stress before encoding change how vividly a memory is later described in language?
- Does bilingualism change how quickly emotional words are recognized and remembered compared to words in a first language?
- Does a brief nap improve retention of newly learned vocabulary compared to an equivalent period of wakefulness?
- Does sleep loss reduce accuracy in recognizing others' facial expressions of emotion?
- Does reading emotionally arousing text before bed alter dream content or later memory for that text?

2. Write a 400-500 word response that does the following:

a. Study Design:

- i. State your hypothesis and predicted direction of effect.
- ii. Identify your experimental and control conditions.

- iii. Describe a realistic experimental method: who and what species your participants would be, what task or stimuli you'd use, how you would measure brain activity (if you plan to), and what outcomes you would measure. Identify at least one alternative explanation your design would need to rule out, and briefly explain how.
 - b. Predicted Results & Interpretation: Describe the results that would support your hypothesis and the results that would challenge it. Explain what each outcome would tell us about how sleep, emotion, memory, or language work.
3. Ground your hypothesis in at least one source beyond course materials (a peer-reviewed study, textbook chapter, or reputable science publication) that informed your thinking. Any citation format is fine.

A Note on AI Use: Using an LLM to help you brainstorm questions, think through experimental design, or track down relevant studies is a great use of AI for this assignment. However, you should not use the LLM to generate your hypothesis, design, or write-up for you — the reasoning behind your design choices should be your own. If you're unsure where the line is, feel free to ask.

Rubric (5 points total)

2 points: Study design is specific, internally consistent, and identifies a real confound with a plausible control

2 points: Predicted results are clearly tied to the hypothesis, and the interpretation reflects accurate neuroscience

1 point: A source grounding the hypothesis is cited, word count is met